CASE STUDY 1 : Student Academic Dashboard

1. Problem Statement

To develop a Student Academic Dashboard website that displays student details, semester-wise results, attendance percentages, subject information, examination schedules, and important announcements. The dashboard should include summary cards, tables, progress bars, and a sidebar navigation menu to organize academic information effectively. The website should provide a responsive, interactive, and user-friendly interface using HTML, CSS, and JavaScript.

2. Objectives

1. To design a user-friendly dashboard for displaying student details and academic information.

2. To display semester-wise examination results and subject-wise marks in an organized table.

3. To represent student attendance percentages using progress bars.

4. To provide subject information, including subject names, codes, and faculty details.

5. To display examination schedules with dates, times, and subject details.

6. To show important academic announcements and notifications.

7. To organize dashboard sections using summary cards and a sidebar navigation menu.

8. To develop a responsive website using HTML, CSS, and JavaScript that works on desktops, tablets, and mobile devices.

3. Assumptions and Business Rules

Assumptions

1. Each student has a unique student ID or roll number.
2. Student details, semester results, attendance, and subject information are available in the system.
3. Semester results include subject names, marks obtained, maximum marks, and grades.

4. Attendance percentages are calculated using the number of classes attended and the total number of classes conducted.
5. Examination schedules include exam dates, times, and subject names.
6. Announcements are provided by the college or academic department.
7. The dashboard is designed to work on desktops, tablets, and mobile devices.
8. Sample student data may be used for demonstration purposes.

Business Rules

1. Every student must have a unique roll number.
2. Attendance percentage is calculated using the formula:

   Attendance Percentage = (Classes Attended / Total Classes) × 100

3. Marks obtained must not exceed the maximum marks for a subject.

4. Semester results must be displayed subject-wise in a structured table.

5. Attendance percentages must be displayed using progress bars.

6. Summary cards must display important academic information, such as average marks, attendance percentage, and total subjects.

7. Examination schedules must display valid dates, times, and subject details.

8. Important announcements must be displayed clearly so students can identify academic updates.

9. The sidebar must provide navigation to the dashboard's different sections.

10. The interface must be responsive and present information clearly across supported screen sizes.

3. Algorithm

1. Start the program.

2. Create the basic webpage structure using HTML.

3. Design the dashboard layout using CSS with a sidebar navigation menu and main content area.

4. Display student details such as name, roll number, department, year, and semester.

5. Create summary cards to show academic performance, total subjects, average marks, and attendance percentage.

6. Display semester-wise results and subject information in tables.

7. Calculate and display attendance percentages using progress bars.

8. Display the examination schedule with exam dates, times, and subject names.

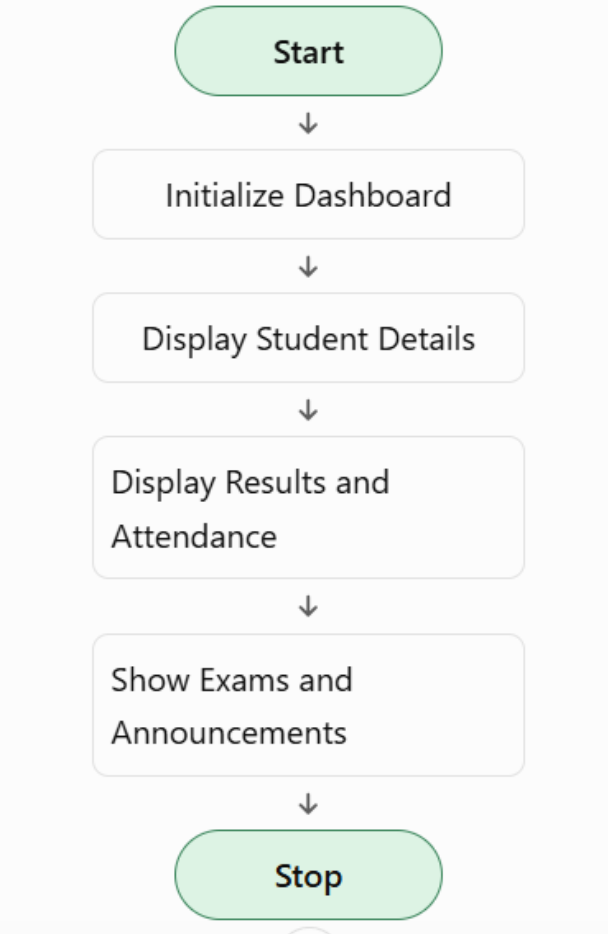
9. Show important announcements and academic notifications.

10. Use JavaScript to implement interactive features such as sidebar navigation and dynamic data display.

11. Apply responsive design to ensure the dashboard works on desktops, tablets, and mobile devices.

12. Stop the program.

4. Flowchart



5. Pseudocode

START

1. Initialize Student Academic Dashboard

2. Display Sidebar Navigation

3. Display Student Details
   Read student name, roll number, department, semester

4. Display Academic Summary Cards
   Show total subjects, average marks, attendance percentage

5. Display Semester Results
   Show subject names, marks, and grades in a table

6. Display Subject Information
   Show subject names and faculty details

7. Calculate Attendance Percentage
   Attendance = (Classes Attended / Total Classes) * 100
   Display attendance using progress bars

8. Display Examination Schedule
   Show exam dates, times, and subjects

9. Display Important Announcements

10. Apply Responsive Design
    Adjust layout for mobile, tablet, and desktop

STOP

6. Code

```
<!-- Sidebar -->
<aside class="sidebar" id="sidebar">
    <div class="logo">🎓 Student Portal</div>

    <p>Main Menu</p>
    <a href="#dashboard" class="active">🏠 Dashboard</a>
    <a href="#profile">👤 Student Profile</a>
    <a href="#results">📊 Semester Results</a>
    <a href="#attendance">📅 Attendance</a>
    <a href="#subjects">📚 Subjects</a>
    <a href="#exams">📝 Exam Schedule</a>
    <a href="#announcements">📢 Announcements</a>
</aside>

<!-- Main -->
<main class="main">

    <div class="topbar" id="dashboard">
        <div style="display:flex;align-items:center;gap:14px;">
            <button class="menu-btn" id="menuBtn"
```

```
                aria-label="Toggle sidebar"
                aria-expanded="false">☰</button>
        <div>
            <h1>Student Dashboard</h1>
            <p>Welcome back! Here is your academic overview.</p>
        </div>
    </div>

    <div class="student-badge">👨‍🎓 B.Tech Student</div>
</div>

<!-- Summary Cards -->
<div class="card-grid">
    <div class="card">
        <div class="stat-icon">📚</div>
        <h3>Current Semester</h3>
        <div class="stat-value">4th</div>
        <div class="stat-note">Academic Year 2026–27</div>
    </div>

    <div class="card">
        <div class="stat-icon">📈</div>
        <h3>Overall CGPA</h3>
        <div class="stat-value">8.45</div>
        <div class="stat-note">Sample academic record</div>
    </div>

    <div class="card">
        <div class="stat-icon">✅</div>
        <h3>Overall Attendance</h3>
        <div class="stat-value">86%</div>
        <div class="stat-note">Above 75% target</div>
    </div>

    <div class="card">
        <div class="stat-icon">📝</div>
        <h3>Upcoming Exams</h3>
        <div class="stat-value">3</div>
```

```
            <div class="stat-note">Check examination schedule</div>
        </div>
    </div>

    <!-- Student Profile -->
    <section class="section" id="profile">
        <div class="section-heading">
            <h2>Student Information</h2>
            <button class="btn" onclick="window.print()">Print
Details</button>
        </div>

        <div class="panel">
            <div class="profile-grid">
                <div class="profile-item">
                    <span>Student Name</span>
                    <strong>Lekhana P.</strong>
                </div>

                <div class="profile-item">
                    <span>Roll Number</span>
                    <strong>23ABC101</strong>
                </div>

                <div class="profile-item">
                    <span>Department</span>
                    <strong>Computer Science</strong>
                </div>

                <div class="profile-item">
                    <span>Course</span>
                    <strong>B.Tech</strong>
                </div>

                <div class="profile-item">
                    <span>Semester</span>
                    <strong>4th Semester</strong>
                </div>
```

```
                <div class="profile-item">
                    <span>Email</span>
                    <strong>student@example.com</strong>
                </div>
            </div>
        </div>
    </section>

    <!-- Semester Results -->
    <section class="section" id="results">
        <div class="section-heading">
            <h2>Semester Results</h2>
        </div>

        <div class="panel">
            <table>
                <thead>
                    <tr>
                        <th>Semester</th>
                        <th>Subjects</th>
                        <th>Percentage</th>
                        <th>SGPA</th>
                        <th>Status</th>
                    </tr>
                </thead>

                <tbody>
                    <tr>
                        <td>Semester 1</td>
                        <td>6</td>
                        <td>82%</td>
                        <td>8.2</td>
                        <td><span class="badge
pass">Passed</span></td>
                    </tr>

                    <tr>
                        <td>Semester 2</td>
                        <td>6</td>
```

```
                        <td>85%</td>
                        <td>8.5</td>
                        <td><span class="badge
pass">Passed</span></td>
                    </tr>

                    <tr>
                        <td>Semester 3</td>
                        <td>7</td>
                        <td>84%</td>
                        <td>8.4</td>
                        <td><span class="badge
pass">Passed</span></td>
                    </tr>

                    <tr>
                        <td>Semester 4</td>
                        <td>6</td>
                        <td>--</td>
                        <td>--</td>
                        <td><span class="badge upcoming">In
Progress</span></td>
                    </tr>
                </tbody>
            </table>
        </div>
    </section>

    <!-- Attendance -->
    <section class="section" id="attendance">
        <div class="section-heading">
            <h2>Subject-wise Attendance</h2>
        </div>

        <div class="panel">
            <div class="attendance-list">

                <div class="attendance-item">
                    <h4>Java Programming</h4>
```

```
                    <div class="progress-info">
                        <span>34 / 40 classes</span>
                        <strong>85%</strong>
                    </div>
                    <div class="progress-bar">
                        <div class="progress-fill green"
style="width:85%"></div>
                    </div>
                </div>

                <div class="attendance-item">
                    <h4>Database Management</h4>
                    <div class="progress-info">
                        <span>36 / 40 classes</span>
                        <strong>90%</strong>
                    </div>
                    <div class="progress-bar">
                        <div class="progress-fill blue"
style="width:90%"></div>
                    </div>
                </div>

                <div class="attendance-item">
                    <h4>Computer Networks</h4>
                    <div class="progress-info">
                        <span>30 / 40 classes</span>
                        <strong>75%</strong>
                    </div>
                    <div class="progress-bar">
                        <div class="progress-fill orange"
style="width:75%"></div>
                    </div>
                </div>

                <div class="attendance-item">
                    <h4>Mathematics</h4>
                    <div class="progress-info">
                        <span>25 / 40 classes</span>
                        <strong>62.5%</strong>
```

```
                    </div>
                    <div class="progress-bar">
                        <div class="progress-fill red"
style="width:62.5%"></div>
                    </div>
                </div>

            </div>
            <p style="margin-top:22px;color:#64748b;font-size:13px;">
                Note: Attendance values are examples. Check your
institution's rules.
            </p>
        </div>
    </section>

    <!-- Subjects -->
    <section class="section" id="subjects">
        <div class="section-heading">
            <h2>Current Semester Subjects</h2>
        </div>

        <div class="panel">
            <table>
                <thead>
                    <tr>
                        <th>Subject Code</th>
                        <th>Subject Name</th>
                        <th>Faculty</th>
                        <th>Credits</th>
                    </tr>
                </thead>

                <tbody>
                    <tr>
                        <td>CS401</td>
                        <td>Java Programming</td>
                        <td>Mr. Ravi Kumar</td>
                        <td>4</td>
                    </tr>
```

```
                <tr>
                    <td>CS402</td>
                    <td>Database Management Systems</td>
                    <td>Mrs. Priya</td>
                    <td>4</td>
                </tr>

                <tr>
                    <td>CS403</td>
                    <td>Computer Networks</td>
                    <td>Mr. Arun</td>
                    <td>3</td>
                </tr>

                <tr>
                    <td>MA404</td>
                    <td>Engineering Mathematics</td>
                    <td>Dr. Suresh</td>
                    <td>4</td>
                </tr>

                <tr>
                    <td>CS405</td>
                    <td>Web Technologies Lab</td>
                    <td>Mrs. Divya</td>
                    <td>2</td>
                </tr>
            </tbody>
        </table>
    </div>
</section>

<!-- Exam Schedule -->
<section class="section" id="exams">
    <div class="section-heading">
        <h2>Examination Schedule</h2>
    </div>
```

```
        <div class="panel">
            <table>
                <thead>
                    <tr>
                        <th>Date</th>
                        <th>Subject</th>
                        <th>Time</th>
                        <th>Exam Type</th>
                        <th>Status</th>
                    </tr>
                </thead>

                <tbody>
                    <tr>
                        <td>10 Nov 2026</td>
                        <td>Java Programming</td>
                        <td>9:30 AM</td>
                        <td>Mid Examination</td>
                        <td><span class="badge
upcoming">Upcoming</span></td>
                    </tr>

                    <tr>
                        <td>12 Nov 2026</td>
                        <td>Database Management</td>
                        <td>9:30 AM</td>
                        <td>Mid Examination</td>
                        <td><span class="badge
upcoming">Upcoming</span></td>
                    </tr>

                    <tr>
                        <td>14 Nov 2026</td>
                        <td>Computer Networks</td>
                        <td>1:30 PM</td>
                        <td>Mid Examination</td>
                        <td><span class="badge
upcoming">Upcoming</span></td>
                    </tr>
```

```
                </tbody>
            </table>
        </div>
    </section>

    <!-- Announcements -->
    <section class="section" id="announcements">
        <div class="section-heading">
            <h2>Announcements</h2>
        </div>

        <div class="two-column">
            <div class="panel">
                <h3>📢 Academic Notices</h3>

                <div class="announcement">
                    <span class="badge info">Academic</span>
                    <h4>Internal Examination Preparation</h4>
                    <p>Students should review their syllabus and
examination guidelines.</p>
                    <small>Sample notice</small>
                </div>

                <div class="announcement">
                    <span class="badge pass">Event</span>
                    <h4>Technical Project Exhibition</h4>
                    <p>Students can prepare projects for the upcoming
department exhibition.</p>
                    <small>Sample notice</small>
                </div>

                <div class="announcement">
                    <span class="badge upcoming">Reminder</span>
                    <h4>Attendance Review</h4>
                    <p>Check subject-wise attendance and contact
faculty about discrepancies.</p>
                    <small>Sample notice</small>
                </div>
            </div>
```

```
            <div class="panel">
                <h3>📌 Important Information</h3>

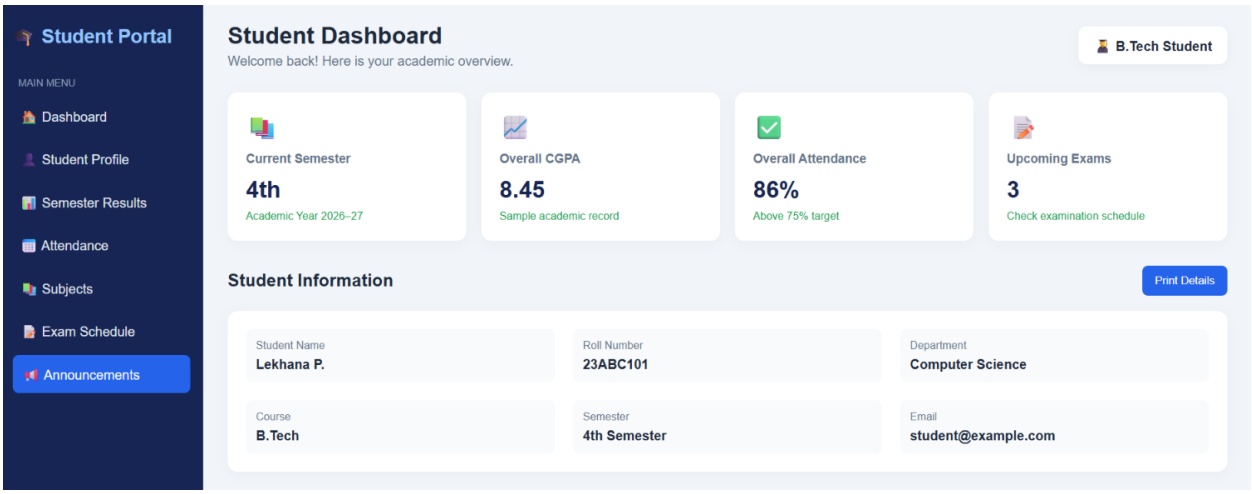
                <div class="announcement">
                    <h4>Library</h4>
                    <p>Return borrowed books by their due dates.</p>
                </div>

                <div class="announcement">
                    <h4>Student Support</h4>
                    <p>Contact your faculty advisor for academic
assistance.</p>
                </div>

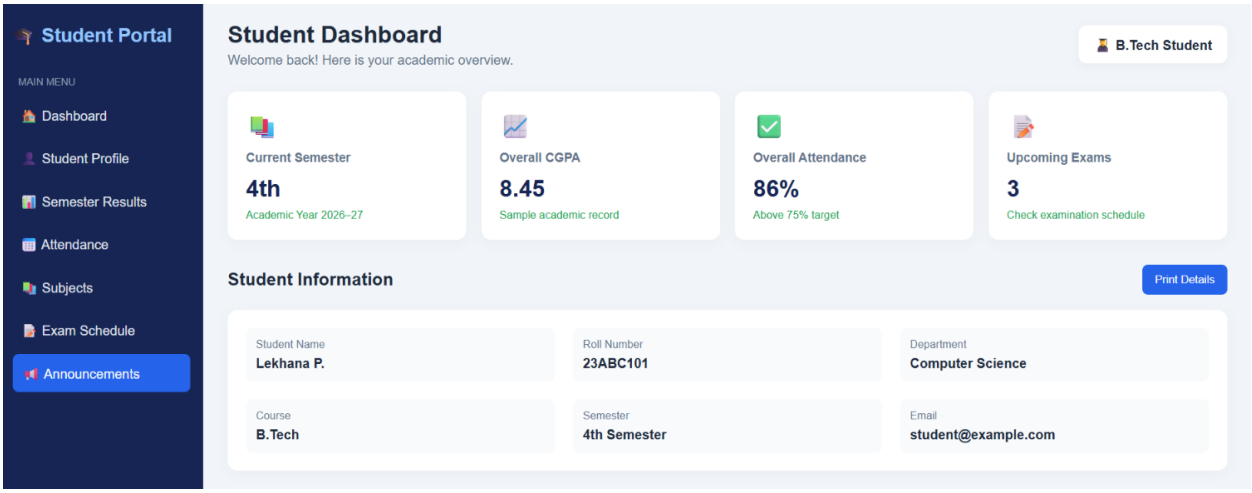
                <div class="announcement">
                    <h4>Exam Instructions</h4>
                    <p>Bring your college identity card to
examinations.</p>
                </div>
            </div>
        </div>
    </section>

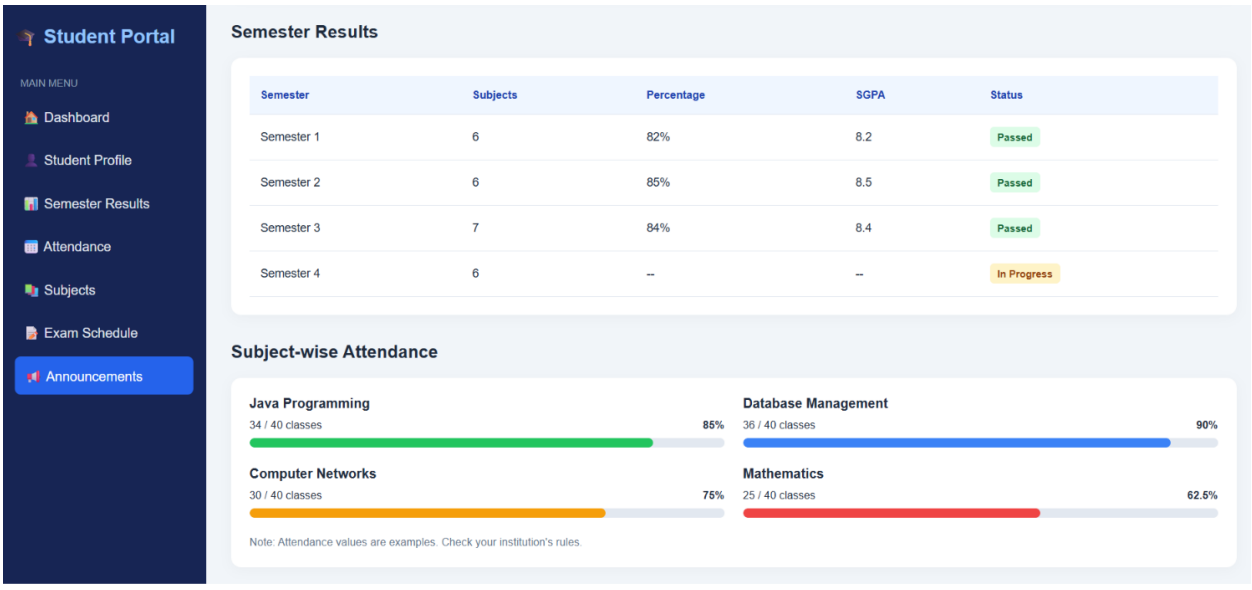
    <footer>
        <p>© 2026 Student Dashboard | Academic Portal Demo</p>
        <p>Designed using HTML, CSS and JavaScript</p>
    </footer>
```

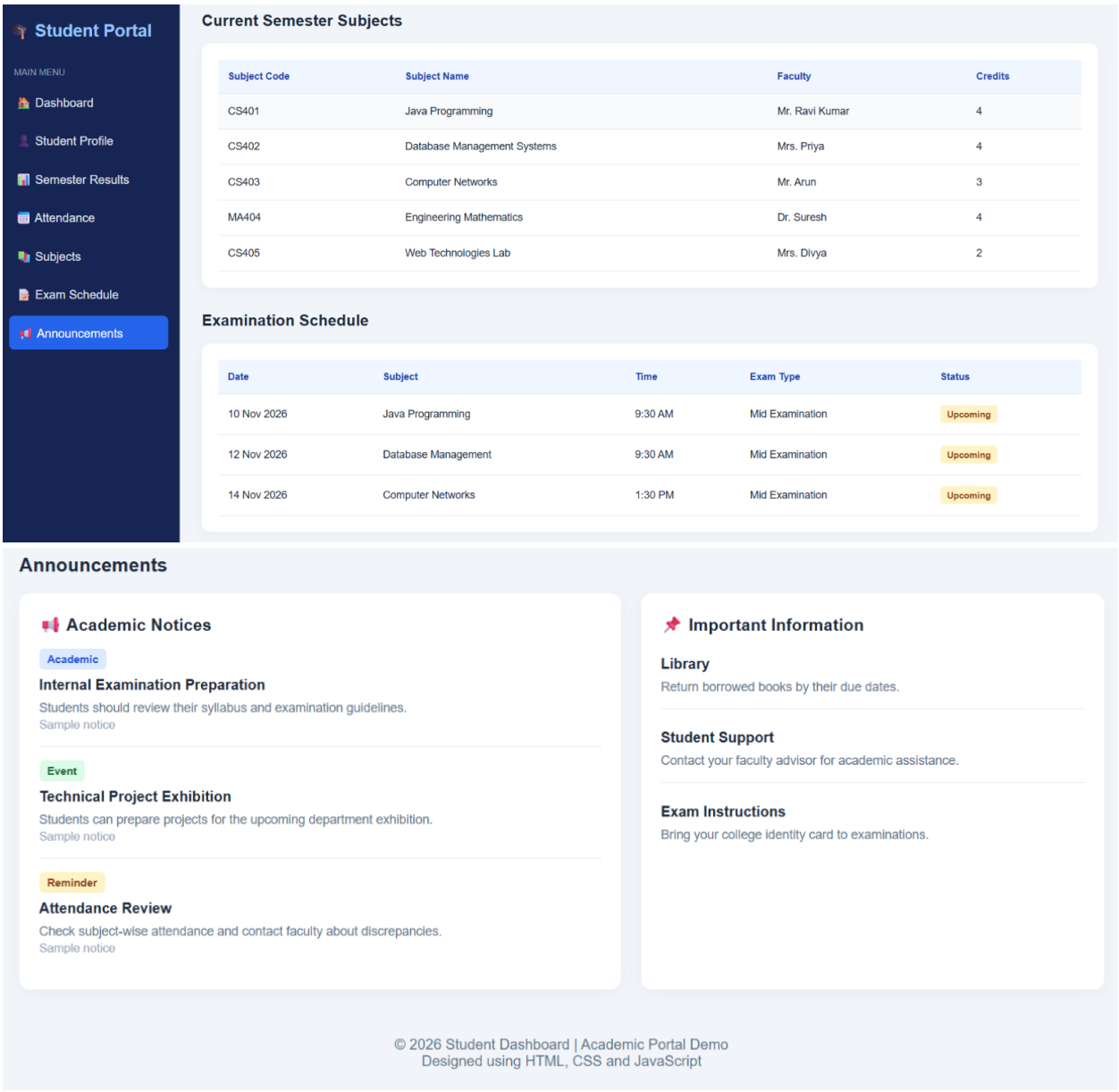
```
</main>
```



7. Output Screenshot







CASE STUDY 2 : Online Shopping Website

1. Problem Statement

To design and develop a responsive e-commerce website that displays products with their prices, discounts, ratings, categories, and descriptions. The website should include a responsive product grid, promotional banners, and a shopping navigation bar. It should provide interactive shopping buttons such as “Add to Cart” and “Buy Now” to improve the user experience. The website should be developed using HTML, CSS, and JavaScript with a user-friendly interface that works efficiently on desktops, tablets, and mobile devices.

2. Objectives

1. To design a user-friendly and responsive e-commerce website.

2. To display products with their names, prices, discounts, ratings, and descriptions.

3. To organize products into different categories for easy browsing.

4. To design a responsive product grid that adapts to different screen sizes.

5. To create attractive promotional banners to highlight special offers and discounts.

6. To develop a shopping navigation bar for easy access to different website sections.

7. To provide interactive shopping buttons such as Add to Cart and Buy Now.

8. To use HTML, CSS, and JavaScript to create an attractive, interactive, and mobile-friendly shopping interface.

3. Assumptions and Business Rules

1. Assumptions

1. The website displays products from different categories.

2. Each product has a name, image, price, discount, rating, and description.

3. Product information is available and displayed correctly.

4. Customers can browse products using the navigation bar and categories.

5. Promotional banners display special offers and discounts.
6. The website is designed for desktop, tablet, and mobile devices.
7. Sample product data is used for demonstration purposes.

2. Business Rules

1. Every product must have a unique product ID.
2. Product prices must be greater than zero.

3. Discount percentages must be between 0% and 100%.
4. The final price is calculated using:

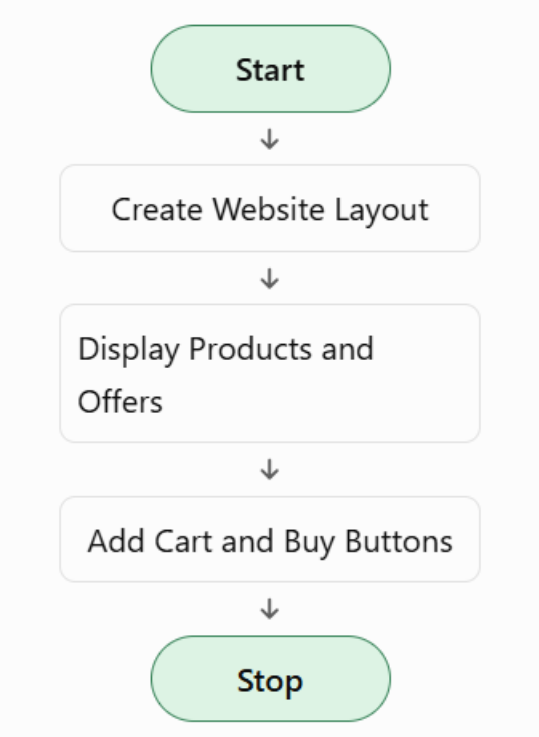
   Final Price = Original Price − Discount Amount

5. Product ratings must be displayed on a scale of 1 to 5.
6. Products must be organized into appropriate categories.
7. The Add to Cart button should allow customers to add products to their cart.
8. The Buy Now button should initiate the purchase process.
9. Promotional banners must display valid offers and applicable discount details.
10. The product grid and navigation bar must adjust to different screen sizes.

4.Algorithm

- Start the program.
- Create the webpage using HTML.
- Design the navigation bar and promotional banners using CSS.
- Display products with prices, discounts, ratings, and descriptions.
- Organize products into categories using a responsive grid.
- Add to Cart and Buy Now buttons using JavaScript.
- Make the website responsive for all devices.
- Stop the program.

5. Flowchart



6. Pseudocode

START

1. Initialize E-Commerce Website

2. Display Navigation Bar and Promotional Banners

3. Display Product Categories

4. For each product:
   Display Name, Price, Discount, Rating
   Display Product Description

5. Display Products in Responsive Grid

6. Provide Add to Cart and Buy Now Buttons

7. Update Cart when a product is added

8. Adjust Layout for Different Devices

STOP

7. **Code**

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ShopEase - Online Shopping</title>

<style>
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background: #f5f7fb;
  color: #222;
}

.navbar {
  background: #172554;
  color: white;
  padding: 18px 6%;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 20px;
  flex-wrap: wrap;
}

.logo {
  font-size: 28px;
  font-weight: bold;
  color: #facc15;
}
```

```
.navbar a {
  text-decoration: none;
  color: white;
  margin: 0 10px;
}

.navbar a:hover {
  color: #facc15;
}

.search {
  display: flex;
  flex: 1;
  max-width: 400px;
}

.search input {
  width: 100%;
  min-width: 0;
  padding: 11px;
  border: none;
  border-radius: 6px 0 0 6px;
}

.search button {
  padding: 11px 15px;
  background: #facc15;
  border: none;
  cursor: pointer;
  border-radius: 0 6px 6px 0;
}

.cart {
  font-weight: bold;
  cursor: pointer;
}
```

```
.hero {
  background: linear-gradient(120deg, #312e81, #7c3aed);
  color: white;
  padding: 65px 8%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 25px;
}

.hero h1 {
  font-size: 42px;
  margin-bottom: 15px;
}

.hero p {
  margin-bottom: 22px;
  font-size: 17px;
}

.hero button {
  background: #facc15;
  border: none;
  padding: 13px 25px;
  border-radius: 6px;
  font-weight: bold;
  cursor: pointer;
}

.hero-art {
  font-size: 110px;
}

.section {
  width: 90%;
  max-width: 1250px;
  margin: 38px auto;
}
```

```
.section h2 {
  margin-bottom: 22px;
  color: #172554;
}

.categories {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(130px, 1fr));
  gap: 16px;
}

.category {
  padding: 22px 10px;
  text-align: center;
  background: white;
  border-radius: 12px;
  cursor: pointer;
  box-shadow: 0 3px 12px #00000008;
  border: 2px solid transparent;
}

.category:hover {
  border-color: #7c3aed;
}

.category span {
  font-size: 32px;
}

.category h3 {
  margin-top: 9px;
  font-size: 15px;
}

.toolbar {
  display: flex;
  gap: 12px;
```

```
  flex-wrap: wrap;
  margin-bottom: 24px;
}

.toolbar input,
.toolbar select {
  padding: 12px;
  border: 1px solid #ddd;
  border-radius: 7px;
  background: white;
  font-size: 14px;
}

.toolbar input {
  flex: 1;
  min-width: 180px;
}

.toolbar select {
  min-width: 145px;
}

.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(225px, 1fr));
  gap: 22px;
}

.product {
  background: white;
  border-radius: 12px;
  overflow: hidden;
  box-shadow: 0 4px 15px #0000000d;
  transition: transform 0.25s;
}

.product:hover {
  transform: translateY(-5px);
```

```
}

.product-image {
  height: 200px;
  display: flex;
  justify-content: center;
  align-items: center;
  font-size: 85px;
  position: relative;
  background: #eef2ff;
}

.product-image img {
  width: 100%;
  height: 100%;
  object-fit: contain;
  padding: 12px;
}

.discount {
  position: absolute;
  top: 12px;
  left: 12px;
  background: #dc2626;
  color: white;
  padding: 5px 9px;
  font-size: 12px;
  border-radius: 5px;
  font-weight: bold;
}

.product-info {
  padding: 17px;
}

.product-info h3 {
  font-size: 18px;
  margin-bottom: 8px;
```

```
}

.description {
  color: #666;
  font-size: 13px;
  min-height: 38px;
  margin-bottom: 10px;
}

.rating {
  color: #f59e0b;
  margin-bottom: 9px;
  font-size: 14px;
}

.price {
  font-size: 21px;
  color: #172554;
  font-weight: bold;
}

.old-price {
  color: #888;
  text-decoration: line-through;
  font-size: 14px;
  margin-left: 7px;
}

.product-info button {
  width: 100%;
  margin-top: 14px;
  padding: 12px;
  border: none;
  border-radius: 7px;
  background: #172554;
  color: white;
  font-size: 14px;
  font-weight: bold;
```

```
  cursor: pointer;
}

.product-info button:hover {
  background: #7c3aed;
}

#emptyMessage {
  display: none;
  padding: 25px;
  text-align: center;
  color: #666;
}

.benefits {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 18px;
}

.benefit {
  background: white;
  padding: 22px;
  border-radius: 10px;
  text-align: center;
}

.benefit span {
  font-size: 30px;
}

footer {
  background: #172554;
  color: white;
  text-align: center;
  padding: 30px 15px;
  margin-top: 45px;
}
```

```
footer p {
  margin-top: 8px;
  color: #dbeafe;
}

@media(max-width: 700px) {
  .navbar {
    justify-content: center;
  }

  .navbar nav {
    text-align: center;
  }

  .search {
    order: 3;
    flex-basis: 100%;
    max-width: none;
  }

  .hero {
    text-align: center;
    justify-content: center;
    padding: 45px 6%;
  }

  .hero h1 {
    font-size: 32px;
  }

  .hero-art {
    display: none;
  }

  .section {
    width: 92%;
  }
```

```
}
</style>
</head>

<body>

<!-- Navigation -->
<header class="navbar">
  <div class="logo">🛍️ ShopEase</div>

  <nav>
    <a href="#home">Home</a>
    <a href="#categories">Categories</a>
    <a href="#products">Products</a>
  </nav>

  <div class="search">
    <input type="search" id="searchInput"
        placeholder="Search products..."
        oninput="renderProducts()">
    <button onclick="renderProducts()">🔍</button>
  </div>

  <div class="cart" onclick="showCart()">
    🛒 Cart (<span id="cartCount">0</span>)
  </div>
</header>

<!-- Promotional Banner -->
<section class="hero" id="home">
  <div>
    <h1>Big Shopping Days!</h1>
    <p>Discover your favorite products at amazing prices.</p>
    <p><strong>Up to 50% OFF</strong> on selected products.</p>
    <button onclick="goToProducts()">Shop Now →</button>
  </div>
  <div class="hero-art">🎁</div>
```

```
</section>

<main>

<!-- Categories -->
<section class="section" id="categories">
  <h2>Shop by Category</h2>

  <div class="categories">
    <div class="category" onclick="chooseCategory('All')">
      <span>🛍️</span>
      <h3>All Products</h3>
    </div>

    <div class="category" onclick="chooseCategory('Electronics')">
      <span>🎧</span>
      <h3>Electronics</h3>
    </div>

    <div class="category" onclick="chooseCategory('Fashion')">
      <span>👕</span>
      <h3>Fashion</h3>
    </div>

    <div class="category" onclick="chooseCategory('Home')">
      <span>🏠</span>
      <h3>Home</h3>
    </div>

    <div class="category" onclick="chooseCategory('Accessories')">
      <span>⌚</span>
      <h3>Accessories</h3>
    </div>
  </div>
</section>

<!-- Product Listing -->
```

```
<section class="section" id="products">
  <h2> Featured Products</h2>

  <div class="toolbar">
    <input type="search" id="filterSearch"
        placeholder="Find a product..."
        oninput="renderProducts()">

    <select id="categoryFilter" onchange="renderProducts()">
      <option value="All">All Categories</option>
      <option value="Electronics">Electronics</option>
      <option value="Fashion">Fashion</option>
      <option value="Home">Home</option>
      <option value="Accessories">Accessories</option>
    </select>

    <select id="sortFilter" onchange="renderProducts()">
      <option value="default">Sort By</option>
      <option value="low">Price: Low to High</option>
      <option value="high">Price: High to Low</option>
      <option value="rating">Top Rated</option>
    </select>
  </div>

  <div class="product-grid" id="productGrid"></div>
  <p id="emptyMessage">No products found. Try another search.</p>
</section>

<!-- Benefits -->
<section class="section">
  <h2>Why Shop With Us?</h2>

  <div class="benefits">
    <div class="benefit">
      <span> </span>
      <h3>Fast Delivery</h3>
      <p>Convenient delivery to your doorstep.</p>
```

```
      </div>

      <div class="benefit">
        <span> 🔓 </span>
        <h3>Secure Shopping</h3>
        <p>A simple and user-friendly shopping experience.</p>
      </div>

      <div class="benefit">
        <span>💳</span>
        <h3>Easy Payments</h3>
        <p>Payment options can be integrated at checkout.</p>
      </div>
    </div>
  </section>

  </main>

  <footer>
    <h2>🛍️ ShopEase</h2>
    <p>Your everyday shopping destination.</p>
    <p>Contact: support@example.com</p>
    <p>© 2026 ShopEase. All rights reserved.</p>
  </footer>

  <script>
  const products = [
    {
      id: 1,
      name: "Wireless Headphones",
      category: "Electronics",
      price: 1499,
      oldPrice: 2499,
      rating: 4.5,
      icon: " 🎧 ",
      color: "#e0e7ff",
      description: "Comfortable headphones with immersive sound."
```

```
},
{
  id: 2,
  name: "Smart Watch",
  category: "Electronics",
  price: 1999,
  oldPrice: 2999,
  rating: 4.3,
  icon: "⌚",
  color: "#dbeafe",
  description: "Track activity and view daily notifications."
},
{
  id: 3,
  name: "Casual T-Shirt",
  category: "Fashion",
  price: 499,
  oldPrice: 799,
  rating: 4.1,
  icon: "👕",
  color: "#fce7f3",
  description: "Everyday comfort with a modern style."
},
{
  id: 4,
  name: "Running Shoes",
  category: "Fashion",
  price: 1799,
  oldPrice: 2499,
  rating: 4.6,
  icon: "👟",
  color: "#ffedd5",
  description: "Lightweight shoes designed for active days."
},
{
  id: 5,
  name: "Table Lamp",
```

```
    category: "Home",
    price: 699,
    oldPrice: 999,
    rating: 4.0,
    icon: " 💡 ",
    color: "#fef3c7",
    description: "A stylish lamp for your desk or bedroom."
  },
  {
    id: 6,
    name: "Travel Backpack",
    category: "Accessories",
    price: 899,
    oldPrice: 1499,
    rating: 4.4,
    icon: " 🎒 ",
    color: "#dcfce7",
    description: "Spacious backpack for college and travel."
  },
  {
    id: 7,
    name: "Bluetooth Speaker",
    category: "Electronics",
    price: 1299,
    oldPrice: 1999,
    rating: 4.2,
    icon: " 🔊 ",
    color: "#ede9fe",
    description: "Portable speaker with clear audio."
  },
  {
    id: 8,
    name: "Sunglasses",
    category: "Accessories",
    price: 399,
    oldPrice: 599,
    rating: 4.0,
```

```
    icon: "🕶",
    color: "#fee2e2",
    description: "A fashionable accessory for sunny days."
  }
];

let cart = [];

function formatPrice(price) {
  return "₹" + price.toLocaleString("en-IN");
}

function renderProducts() {
  const search =
    (document.getElementById("filterSearch").value ||
     document.getElementById("searchInput").value).toLowerCase();

  const category = document.getElementById("categoryFilter").value;
  const sort = document.getElementById("sortFilter").value;

  let filtered = products.filter(product => {
    const matchesSearch =
      product.name.toLowerCase().includes(search) ||
      product.description.toLowerCase().includes(search) ||
      product.category.toLowerCase().includes(search);

    const matchesCategory =
      category === "All" || product.category === category;

    return matchesSearch && matchesCategory;
  });

  if (sort === "low") filtered.sort((a, b) => a.price - b.price);
  if (sort === "high") filtered.sort((a, b) => b.price - a.price);
  if (sort === "rating") filtered.sort((a, b) => b.rating - a.rating);

  const grid = document.getElementById("productGrid");
```

```
  grid.innerHTML = filtered.map(product => {
    const discount = Math.round(
      (product.oldPrice - product.price) / product.oldPrice * 100
    );

    return `
    <article class="product">
      <div class="product-image" style="background:${product.color}">
        <span class="discount">${discount}% OFF</span>
        ${product.icon}
      </div>

      <div class="product-info">
        <h3>${product.name}</h3>
        <p class="description">${product.description}</p>

        <p class="rating">
          ★★★★★ <span style="color:#555">
          ${product.rating} / 5</span>
        </p>

        <p>
          <span class="price">${formatPrice(product.price)}</span>
          <span class="old-price">${formatPrice(product.oldPrice)}</span>
        </p>

        <button onclick="addToCart(${product.id})">
          Add to Cart 🛒
        </button>
      </div>
    </article>`;
  }).join("");

  document.getElementById("emptyMessage").style.display =
    filtered.length === 0 ? "block" : "none";
}
```

```
function chooseCategory(category) {
  document.getElementById("categoryFilter").value = category;
  document.getElementById("searchInput").value = "";
  document.getElementById("filterSearch").value = "";
  renderProducts();
  goToProducts();
}

function goToProducts() {
  document.getElementById("products").scrollIntoView({
    behavior: "smooth"
  });
}

function addToCart(id) {
  const product = products.find(item => item.id === id);
  cart.push(product);
  document.getElementById("cartCount").textContent = cart.length;
  alert(product.name + " added to your cart!");
}

function showCart() {
  if (cart.length === 0) {
    alert("Your shopping cart is empty.");
    return;
  }

  const counts = {};
  cart.forEach(product => {
    counts[product.name] = (counts[product.name] || 0) + 1;
  });

  const total = cart.reduce((sum, product) => sum + product.price, 0);

  const items = Object.entries(counts)
    .map(([name, count]) => `${name} x ${count}`)
    .join("\n");
```

```
    alert("YOUR CART\n\n" + items +
        "\n\nTotal: " + formatPrice(total) +
        "\n\nDemo only: checkout is not connected.");
}

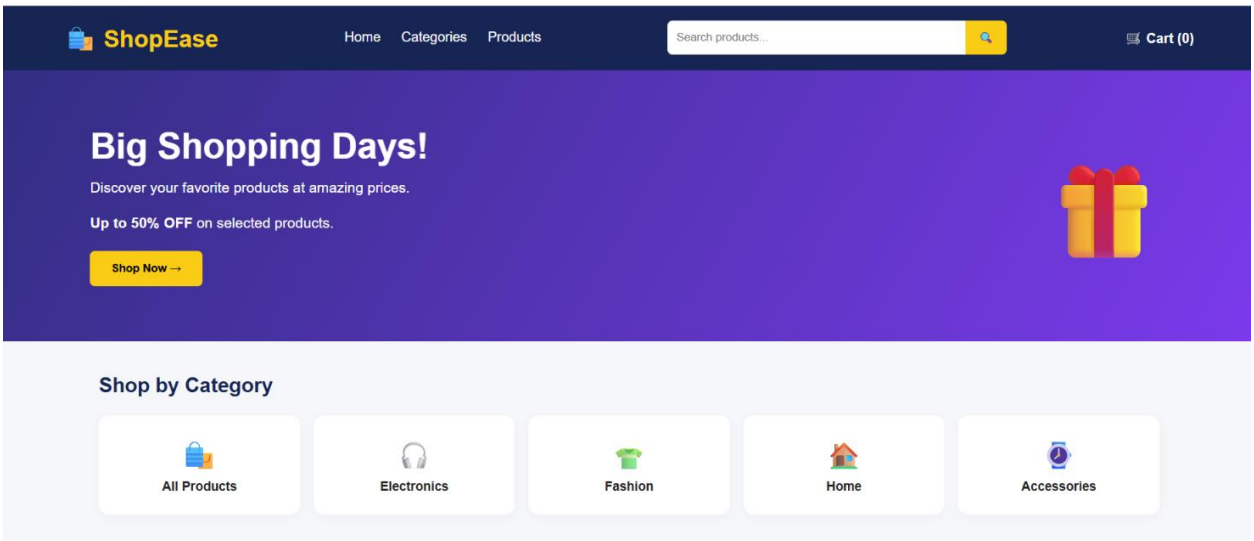
document.getElementById("searchInput").addEventListener("input", function() {
    document.getElementById("filterSearch").value = this.value;
    renderProducts();
});

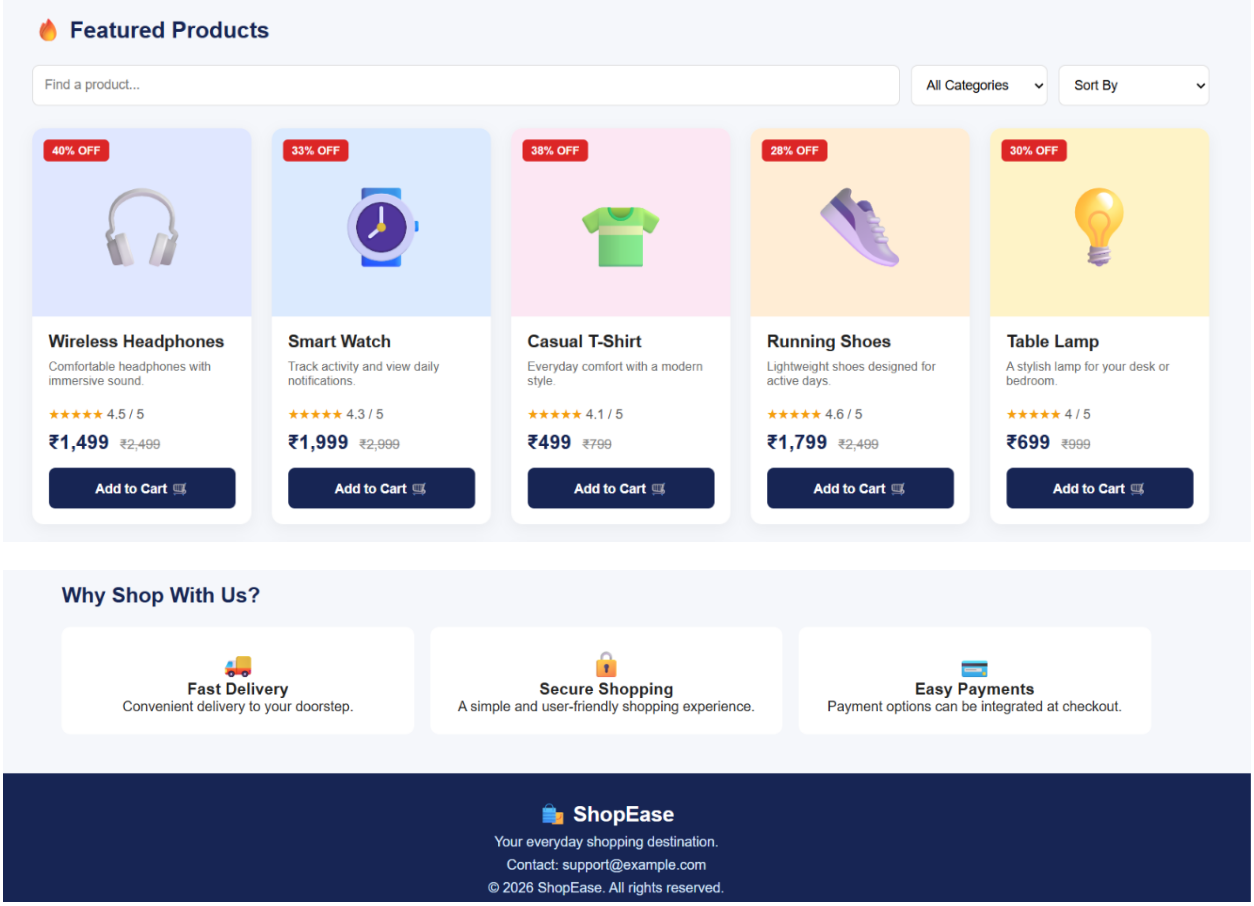
document.getElementById("filterSearch").addEventListener("input", function() {
    document.getElementById("searchInput").value = this.value;
    renderProducts();
});

renderProducts();
</script>

</body>
</html>
```

8. Output Screenshorts





CASE STUDY 3 : Electronics Store Website

1.Problem Statement

To design and develop a responsive online electronics catalog website that displays laptops, smartphones, accessories, and smart devices with their specifications, prices, discounts, and descriptions. The website should include product comparison tables, category filters, discount labels, and detailed product layouts to help users explore and compare electronic products easily. The interface should be user-friendly, interactive, and responsive across desktops, tablets, and mobile devices using HTML, CSS, and JavaScript.

2.Objectives

1. To develop a user-friendly and responsive online electronics catalog website.

2. To display laptops, smartphones, accessories, and smart devices with product details.

3. To provide product specifications such as processor, RAM, storage, battery, and display.

4. To display product prices, discounts, and offers using discount labels.

5. To implement category filters for easy product browsing.

6. To create product comparison tables to compare specifications and prices.

7. To provide detailed product layouts with images, descriptions, and features.

8. To develop an interactive and responsive website using HTML, CSS, and JavaScript.

3.Assumptions and Business Rules

1. Assumptions

1. The website displays laptops, smartphones, accessories, and smart devices.
2. Each product has a name, image, price, specifications, and description.
3. Product prices and discount details are available.
4. Users can filter products by category.
5. Users can compare products using comparison tables.
6. Sample product data is used for demonstration purposes.
7. The website supports desktop, tablet, and mobile devices.

2. Business Rules

1. Each product must have a unique product ID.
2. Product prices must be greater than zero.
3. Discount percentages must be between 0% and 100%.
4. The final price is calculated as:

   Final Price = Original Price − Discount Amount

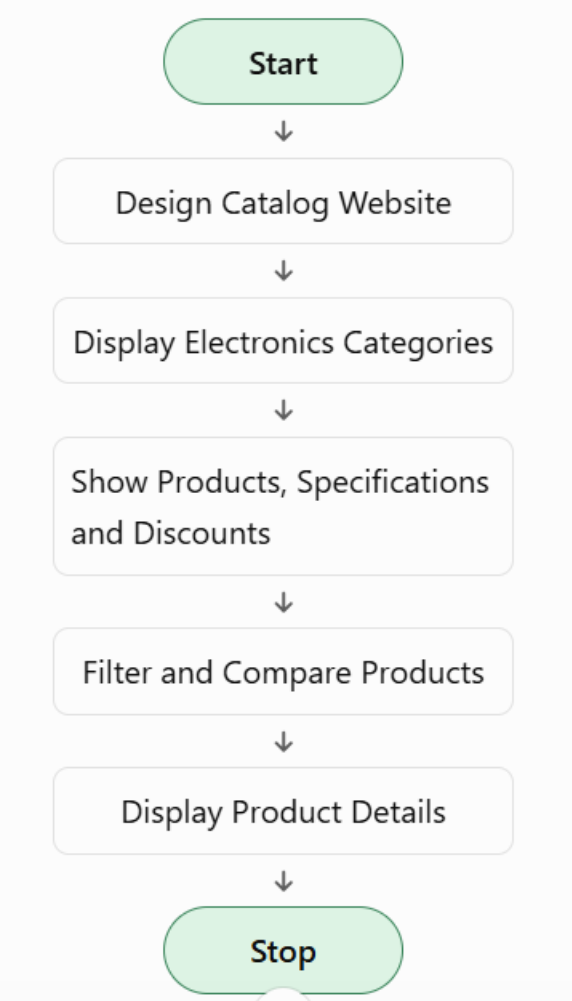
5. Products must be classified into the appropriate categories.
6. Product specifications must be displayed clearly and accurately.
7. Product comparison tables must show the selected products' specifications and prices.
8. Category filters must display only products matching the selected category.
9. Product detail pages must include product images, descriptions, prices, and specifications.

10. The website layout must be responsive and user-friendly across different devices.

4.Algorithm

1. Start the program.
2. Create the website layout using HTML and CSS.
3. Display laptops, smartphones, accessories, and smart devices.
4. Show product specifications, prices, and discount labels.
5. Implement category filters to sort products.
6. Display comparison tables for selected products.
7. Show detailed product information.
8. Make the website responsive using CSS and JavaScript.
9. Stop the program.

5.Flowchart



6.Pseudocode

START

1. Initialize Electronics Catalog

2. Display Categories:

Laptops, Smartphones, Accessories, Smart Devices

3. Display Products with Prices and Discounts

4. Display Product Specifications and Details

5. Apply Category Filters

6. Select Products for Comparison

7. Display Comparison Table

8. Show Detailed Product Information

9. Adjust Layout for Different Devices

STOP

7.Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ElectroMart | Electronics Catalog</title>

<style>
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
```

```
}

body {
  font-family: Arial, sans-serif;
  background: #f4f6fa;
  color: #202b3c;
  line-height: 1.6;
}

header {
  background: #101d36;
  color: white;
  padding: 18px 6%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 15px;
}

.logo {
  font-size: 27px;
  font-weight: bold;
  color: #50d5c5;
}
```

```
nav {
  display: flex;
  gap: 18px;
  flex-wrap: wrap;
}

nav a {
  color: white;
  text-decoration: none;
}

nav a:hover {
  color: #50d5c5;
}

.cart {
  background: #263957;
  padding: 9px 15px;
  border-radius: 8px;
}

.hero {
  background: linear-gradient(120deg, #101d36, #185a79);
  color: white;
  padding: 55px 7%;
  display: flex;
```

```
    align-items: center;
    justify-content: space-between;
    gap: 30px;
}

.hero h1 {
    font-size: 39px;
    margin-bottom: 12px;
}

.hero p {
    color: #e0f2fe;
    margin-bottom: 20px;
}

.hero .hero-icon {
    font-size: 100px;
}

.btn {
    display: inline-block;
    padding: 12px 20px;
    border: none;
    border-radius: 7px;
    background: #50d5c5;
    color: #10253b;
```

```
    font-weight: bold;
    cursor: pointer;
}

.btn:hover {
    background: #2cb5a5;
}

.container {
    width: 90%;
    max-width: 1250px;
    margin: 38px auto;
}

.section-title {
    color: #101d36;
    margin-bottom: 20px;
    font-size: 26px;
}

.category-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
    gap: 18px;
}
```

```
.category-card {
  background: white;
  border-radius: 12px;
  padding: 22px 12px;
  text-align: center;
  cursor: pointer;
  border: 2px solid transparent;
  box-shadow: 0 3px 12px #00000008;
}

.category-card:hover,
.category-card.active {
  border-color: #1b9d91;
}

.category-card span {
  font-size: 36px;
}

.category-card h3 {
  font-size: 16px;
  margin-top: 8px;
}

.toolbar {
  display: flex;
```

```
  flex-wrap: wrap;
  gap: 12px;
  margin-bottom: 25px;
}

.toolbar input,
.toolbar select {
  padding: 12px;
  border: 1px solid #d5dce5;
  border-radius: 7px;
  background: white;
  font-size: 14px;
}

.toolbar input {
  flex: 1;
  min-width: 180px;
}

.toolbar select {
  min-width: 155px;
}

.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(235px, 1fr));
```

```
    gap: 22px;
}

.product-card {
    background: white;
    border-radius: 12px;
    overflow: hidden;
    box-shadow: 0 4px 15px #0000000c;
    transition: 0.25s;
}

.product-card:hover {
    transform: translateY(-5px);
}

.product-image {
    height: 190px;
    background: #eaf0fa;
    display: flex;
    justify-content: center;
    align-items: center;
    font-size: 85px;
    position: relative;
}

.discount {
```

```
  position: absolute;
  left: 12px;
  top: 12px;
  background: #e63946;
  color: white;
  padding: 5px 9px;
  border-radius: 5px;
  font-size: 12px;
  font-weight: bold;
}

.product-info {
  padding: 18px;
}

.product-info h3 {
  margin-bottom: 7px;
  font-size: 18px;
}

.rating {
  color: #e6a000;
  margin: 7px 0;
}

.price {
```

```
    font-size: 22px;
    font-weight: bold;
    color: #102c50;
}

.old-price {
    color: #8992a1;
    text-decoration: line-through;
    font-size: 14px;
    margin-left: 6px;
}

.stock {
    display: inline-block;
    font-size: 12px;
    color: #137443;
    background: #dcfce7;
    padding: 4px 9px;
    border-radius: 20px;
    margin: 8px 0;
}

.product-actions {
    display: flex;
    gap: 8px;
    margin-top: 12px;
```

```
}

.product-actions button {
  flex: 1;
  padding: 10px 5px;
  border-radius: 6px;
  border: none;
  cursor: pointer;
  font-weight: bold;
}

.details-btn {
  background: #e8eef7;
  color: #102c50;
}

.compare-btn {
  background: #102c50;
  color: white;
}

.product-actions button:hover {
  opacity: 0.8;
}

.table-wrap {
```

```
  overflow-x: auto;
  border-radius: 10px;
  background: white;
  box-shadow: 0 3px 14px #00000008;
}

table {
  border-collapse: collapse;
  width: 100%;
  min-width: 680px;
}

th, td {
  padding: 14px;
  text-align: left;
  border-bottom: 1px solid #e5eaf0;
}

th {
  background: #101d36;
  color: white;
}

tbody tr:hover {
  background: #f0fdfa;
}
```

```
.remove-btn {
  background: #fee2e2;
  color: #b91c1c;
  padding: 6px 10px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
}

.clear-btn {
  margin-top: 15px;
  padding: 10px 15px;
  background: #e5eaf2;
  border: none;
  border-radius: 6px;
  cursor: pointer;
}

.spec-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 18px;
}

.spec-card {
```

```
    padding: 20px;
    border-radius: 10px;
    background: white;
    border-left: 4px solid #1b9d91;
}

.spec-card h3 {
    margin-bottom: 8px;
}

#detailPanel {
    display: none;
    background: white;
    border-radius: 12px;
    padding: 25px;
    margin-top: 20px;
    box-shadow: 0 4px 16px #0000000c;
}

.detail-layout {
    display: grid;
    grid-template-columns: 220px 1fr;
    gap: 25px;
    align-items: start;
}
```

```
.detail-image {
  min-height: 190px;
  border-radius: 10px;
  background: #eaf0fa;
  display: flex;
  justify-content: center;
  align-items: center;
  font-size: 95px;
}

.detail-specs {
  margin: 12px 0;
}

.detail-specs p {
  padding: 7px 0;
  border-bottom: 1px solid #eee;
}

#noResults {
  display: none;
  text-align: center;
  padding: 25px;
}

footer {
```

```
    background: #101d36;
    color: white;
    text-align: center;
    padding: 30px 15px;
    margin-top: 45px;
}

footer p {
    color: #cbd5e1;
    margin-top: 7px;
}

@media(max-width: 650px) {
    header {
        justify-content: center;
    }

    .hero {
        text-align: center;
        justify-content: center;
    }

    .hero h1 {
        font-size: 30px;
    }
```

```
    .hero-icon {
        display: none;
    }

    .detail-layout {
        grid-template-columns: 1fr;
    }

    .container {
        width: 93%;
    }
}
</style>
</head>

<body>

<header>
    <div class="logo">⚡ ElectroMart</div>

    <nav>
        <a href="#home">Home</a>
        <a href="#categories">Categories</a>
        <a href="#products">Products</a>
        <a href="#comparison">Compare</a>
    </nav>
```

```
    <div class="cart">
      🛒 Cart: <span id="cartCount">0</span>
    </div>
  </header>

  <section class="hero" id="home">
    <div>
      <h1>Upgrade Your Technology</h1>
      <p>Explore laptops, smartphones, accessories and smart devices.</p>
      <p><strong>Special deals on selected electronics!</strong></p>
      <button class="btn" onclick="scrollToProducts()">Explore Products →</button>
    </div>

    <div class="hero-icon">💻</div>
  </section>

  <main class="container">

    <section id="categories">
      <h2 class="section-title">Browse Categories</h2>

      <div class="category-grid">
        <div class="category-card active" data-category="All"
          onclick="selectCategory('All')">
          <span>🛍️</span>
```

```
  <h3>All Products</h3>
</div>

<div class="category-card" data-category="Laptops"
   onclick="selectCategory('Laptops')">
  <span>💻</span>
  <h3>Laptops</h3>
</div>

<div class="category-card" data-category="Smartphones"
   onclick="selectCategory('Smartphones')">
  <span> 📱 </span>
  <h3>Smartphones</h3>
</div>

<div class="category-card" data-category="Accessories"
   onclick="selectCategory('Accessories')">
  <span> 🎧 </span>
  <h3>Accessories</h3>
</div>

<div class="category-card" data-category="Smart Devices"
   onclick="selectCategory('Smart Devices')">
  <span> ⌚ </span>
  <h3>Smart Devices</h3>
</div>
```

```
  </div>
</section>

<section id="products" style="margin-top:42px">
  <h2 class="section-title">Featured Electronics</h2>

  <div class="toolbar">
    <input type="search" id="searchInput"
        placeholder="Search products or specifications..."
        oninput="renderProducts()">

    <select id="categoryFilter" onchange="syncCategory();renderProducts()">
      <option value="All">All Categories</option>
      <option value="Laptops">Laptops</option>
      <option value="Smartphones">Smartphones</option>
      <option value="Accessories">Accessories</option>
      <option value="Smart Devices">Smart Devices</option>
    </select>

    <select id="sortFilter" onchange="renderProducts()">
      <option value="default">Sort Products</option>
      <option value="low">Price: Low to High</option>
      <option value="high">Price: High to Low</option>
      <option value="rating">Highest Rated</option>
      <option value="discount">Biggest Discount</option>
    </select>
```

```
  </div>

  <div class="product-grid" id="productGrid"></div>
  <p id="noResults">No matching products found.</p>
</section>

<section id="productDetails" style="margin-top:42px">
  <h2 class="section-title">Product Details & Specifications</h2>

  <p>Select "View Details" on any product to see its specifications.</p>

  <div id="detailPanel"></div>
</section>

<section id="comparison" style="margin-top:45px">
  <h2 class="section-title">📊 Product Comparison</h2>

  <p style="margin-bottom:15px">
    Select "Compare" on product cards to add products to this table.
    Compare up to 4 products.
  </p>

  <div class="table-wrap">
    <table>
      <thead>
        <tr>
```

```
            <th>Product</th>
            <th>Category</th>
            <th>Processor / Feature</th>
            <th>Memory / Storage</th>
            <th>Display</th>
            <th>Price</th>
            <th>Rating</th>
            <th>Action</th>
          </tr>
        </thead>

        <tbody id="comparisonBody">
          <tr>
            <td colspan="8">No products selected. Add products using Compare.</td>
          </tr>
        </tbody>
      </table>
    </div>

    <button class="clear-btn" onclick="clearComparison()">
      Clear Comparison
    </button>
  </section>

  <section style="margin-top:45px">
    <h2 class="section-title">Why Shop at ElectroMart?</h2>
```

```
      <div class="spec-grid">
        <div class="spec-card">
          <h3>🚚 Reliable Delivery</h3>
          <p>Convenient delivery options for your electronics.</p>
        </div>

        <div class="spec-card">
          <h3>🌓 Warranty Information</h3>
          <p>Check the manufacturer's warranty for each product.</p>
        </div>

        <div class="spec-card">
          <h3>💳 Easy Shopping</h3>
          <p>Browse products and compare specifications in one place.</p>
        </div>
      </div>
    </section>

</main>

<footer>
    <h2> ⚡ ElectroMart</h2>
    <p>Your destination for modern technology.</p>
    <p>Email: support@electromart.example</p>
    <p>© 2026 ElectroMart. All rights reserved.</p>
```

```
</footer>

<script>
const products = [
  {
    id: 1,
    name: "Performance Laptop",
    category: "Laptops",
    price: 54999,
    oldPrice: 69999,
    rating: 4.6,
    icon: "💻",
    processor: "Intel Core i5",
    memory: "16GB RAM / 512GB SSD",
    display: "15.6-inch Full HD",
    battery: "Up to 8 hours",
    description: "A powerful laptop for college, office work and multitasking.",
    color: "#e0e7ff"
  },
  {
    id: 2,
    name: "UltraBook Pro",
    category: "Laptops",
    price: 74999,
    oldPrice: 89999,
    rating: 4.7,
```

```
    icon: "💻",
    processor: "Intel Core i7",
    memory: "16GB RAM / 1TB SSD",
    display: "14-inch Full HD",
    battery: "Up to 10 hours",
    description: "A lightweight laptop for productivity and creative work.",
    color: "#dbeafe"
  },
  {
    id: 3,
    name: "Smartphone X",
    category: "Smartphones",
    price: 24999,
    oldPrice: 29999,
    rating: 4.5,
    icon: "📱",
    processor: "Octa-core processor",
    memory: "8GB RAM / 128GB",
    display: "6.5-inch AMOLED",
    battery: "5000mAh",
    description: "A modern smartphone with a vibrant display and fast performance.",
    color: "#fce7f3"
  },
  {
    id: 4,
    name: "Smartphone Ultra",
```

```
    category: "Smartphones",
    price: 39999,
    oldPrice: 49999,
    rating: 4.8,
    icon: " 📲 ",
    processor: "Flagship octa-core",
    memory: "12GB RAM / 256GB",
    display: "6.7-inch AMOLED",
    battery: "5000mAh",
    description: "A premium smartphone for photography, gaming and everyday use.",
    color: "#ede9fe"
  },
  {
    id: 5,
    name: "Wireless Earbuds",
    category: "Accessories",
    price: 1999,
    oldPrice: 3499,
    rating: 4.3,
    icon: " 🎧 ",
    processor: "Bluetooth 5.3",
    memory: "Charging case included",
    display: "Not applicable",
    battery: "Up to 24 hours with case",
    description: "Compact wireless earbuds for music and calls.",
    color: "#ffedd5"
```

```
},
{
  id: 6,
  name: "Fast Charger",
  category: "Accessories",
  price: 799,
  oldPrice: 1299,
  rating: 4.2,
  icon: " 🔌 ",
  processor: "USB-C Power Delivery",
  memory: "Not applicable",
  display: "Not applicable",
  battery: "Not applicable",
  description: "A compact charger for compatible electronic devices.",
  color: "#dcfce7"
},
{
  id: 7,
  name: "Smart Watch",
  category: "Smart Devices",
  price: 2999,
  oldPrice: 4499,
  rating: 4.4,
  icon: " ⌚ ",
  processor: "Smart wearable chipset",
  memory: "Fitness and activity tracking",
```

```
    display: "1.8-inch touchscreen",
    battery: "Up to 7 days, usage dependent",
    description: "Track activity, monitor workouts and view notifications.",
    color: "#fef3c7"
  },
  {
    id: 8,
    name: "Smart Speaker",
    category: "Smart Devices",
    price: 2499,
    oldPrice: 3999,
    rating: 4.1,
    icon: "🔊",
    processor: "Smart audio processor",
    memory: "Wireless connectivity",
    display: "Not applicable",
    battery: "Power adapter required",
    description: "A compact speaker for listening to music at home.",
    color: "#fee2e2"
  }
];

let comparison = [];
let cart = [];

function money(value) {
```

```
  return "₹" + value.toLocaleString("en-IN");
}

function discount(product) {
  return Math.round(
    (product.oldPrice - product.price) / product.oldPrice * 100
  );
}

function renderProducts() {
  const search = document.getElementById("searchInput")
    .value.toLowerCase().trim();

  const category = document.getElementById("categoryFilter").value;
  const sort = document.getElementById("sortFilter").value;

  let list = products.filter(p => {
    const text = [
      p.name, p.category, p.description,
      p.processor, p.memory, p.display
    ].join(" ").toLowerCase();

    return text.includes(search) &&
      (category === "All" || p.category === category);
  });
```

```
if (sort === "low") list.sort((a,b) => a.price-b.price);
if (sort === "high") list.sort((a,b) => b.price-a.price);
if (sort === "rating") list.sort((a,b) => b.rating-a.rating);
if (sort === "discount") list.sort((a,b) => discount(b)-discount(a));

document.getElementById("productGrid").innerHTML = list.map(p => `
  <article class="product-card">
    <div class="product-image" style="background:${p.color}">
      <span class="discount">${discount(p)}% OFF</span>
      ${p.icon}
    </div>

    <div class="product-info">
      <h3>${p.name}</h3>
      <p style="color:#64748b;font-size:13px">${p.category}</p>
      <p class="rating">★ ${p.rating} / 5</p>
      <p>
        <span class="price">${money(p.price)}</span>
        <span class="old-price">${money(p.oldPrice)}</span>
      </p>
      <span class="stock">● Product listed</span>

      <div class="product-actions">
        <button class="details-btn" onclick="showDetails(${p.id})">
          View Details
        </button>
```

```
            <button class="compare-btn" onclick="addCompare(${p.id})">
              Compare
            </button>
          </div>

          <button class="btn" style="width:100%;margin-top:9px"
            onclick="addCart(${p.id})">
            Add to Cart 🛒
          </button>
        </div>
      </article>
    `).join("");

  document.getElementById("noResults").style.display =
    list.length ? "none" : "block";
}

function selectCategory(category) {
  document.getElementById("categoryFilter").value = category;

  document.querySelectorAll(".category-card").forEach(card => {
    card.classList.toggle(
      "active", card.dataset.category === category
    );
  });
```

```
  renderProducts();
}

function syncCategory() {
  const category = document.getElementById("categoryFilter").value;

  document.querySelectorAll(".category-card").forEach(card => {
    card.classList.toggle(
      "active", card.dataset.category === category
    );
  });
}

function showDetails(id) {
  const p = products.find(item => item.id === id);
  const panel = document.getElementById("detailPanel");

  panel.style.display = "block";
  panel.innerHTML = `
    <div class="detail-layout">
      <div class="detail-image">${p.icon}</div>

      <div>
        <h2>${p.name}</h2>
        <p style="color:#64748b">${p.category}</p>
        <p class="rating">★ ${p.rating} / 5 customer rating (demo)</p>
```

```
        <p>
          <span class="price">${money(p.price)}</span>
          <span class="old-price">${money(p.oldPrice)}</span>
          <span class="discount" style="position:static;display:inline-block">
            ${discount(p)}% OFF
          </span>
        </p>

        <p style="margin-top:12px">${p.description}</p>

        <div class="detail-specs">
          <p><strong>Processor / Connectivity:</strong> ${p.processor}</p>
          <p><strong>Memory / Storage:</strong> ${p.memory}</p>
          <p><strong>Display:</strong> ${p.display}</p>
          <p><strong>Battery / Power:</strong> ${p.battery}</p>
        </div>

        <button class="btn" onclick="addCart(${p.id})">
          Add to Cart
        </button>
      </div>
    </div>
  `;

panel.scrollIntoView({behavior:"smooth", block:"start"});
```

```
}

function addCompare(id) {
  if (comparison.some(p => p.id === id)) {
    alert("This product is already in the comparison table.");
    return;
  }

  if (comparison.length >= 4) {
    alert("You can compare up to 4 products at a time.");
    return;
  }

  comparison.push(products.find(p => p.id === id));
  renderComparison();

  document.getElementById("comparison").scrollIntoView({
    behavior:"smooth"
  });
}

function renderComparison() {
  const body = document.getElementById("comparisonBody");

  if (comparison.length === 0) {
    body.innerHTML = `
```

```
      <tr><td colspan="8">
        No products selected. Add products using Compare.
      </td></tr>`;
    return;
  }

  body.innerHTML = comparison.map(p => `
    <tr>
      <td>${p.icon} ${p.name}</td>
      <td>${p.category}</td>
      <td>${p.processor}</td>
      <td>${p.memory}</td>
      <td>${p.display}</td>
      <td>${money(p.price)}</td>
      <td>★ ${p.rating}</td>
      <td>
        <button class="remove-btn" onclick="removeCompare(${p.id})">
          Remove
        </button>
      </td>
    </tr>
  `).join("");
}

function removeCompare(id) {
  comparison = comparison.filter(p => p.id !== id);
```

```
  renderComparison();
}

function clearComparison() {
  comparison = [];
  renderComparison();
}

function addCart(id) {
  const p = products.find(item => item.id === id);
  cart.push(p);

  document.getElementById("cartCount").textContent = cart.length;
  alert(p.name + " added to your demo cart.");
}

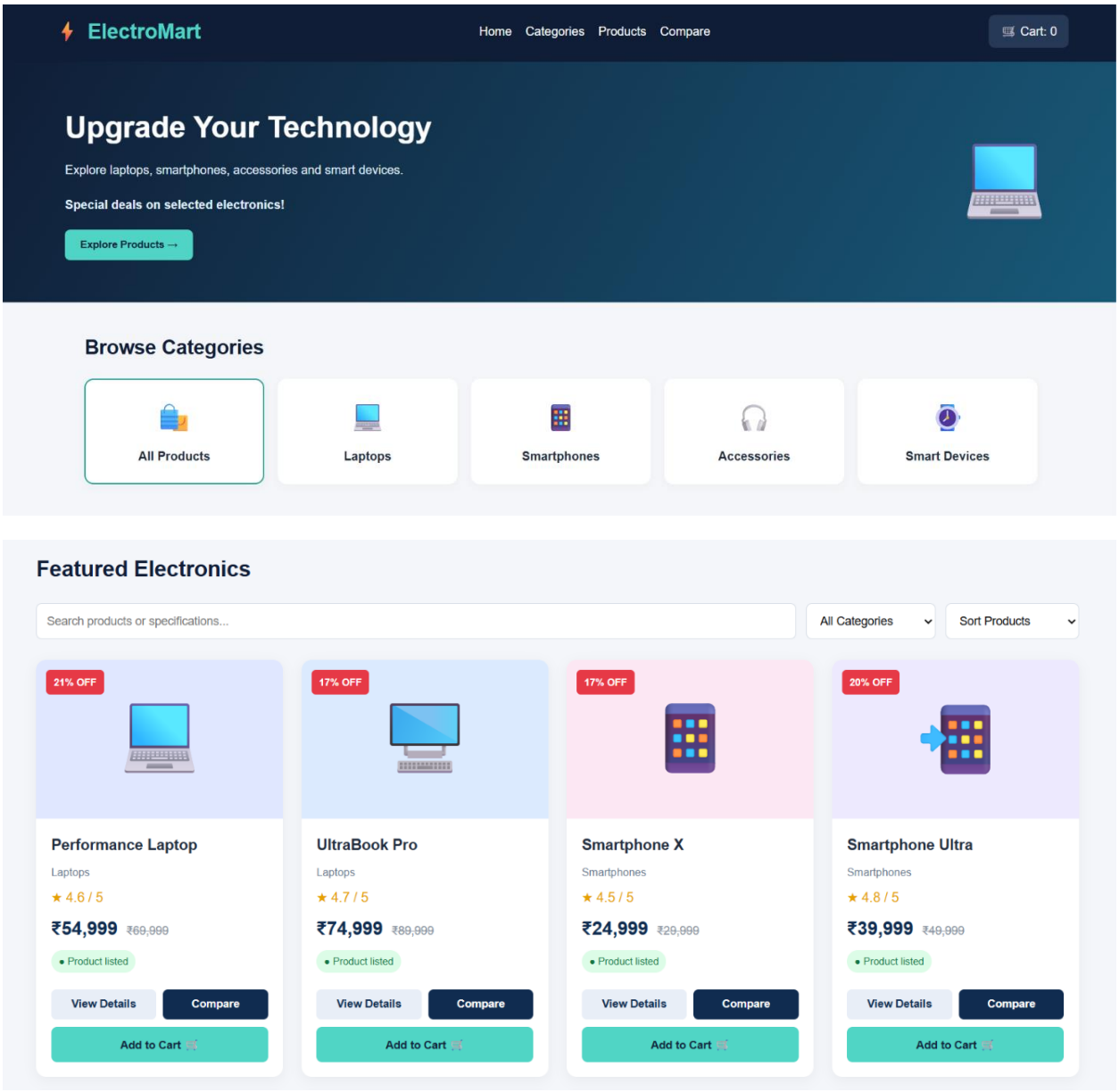
function scrollToProducts() {
  document.getElementById("products").scrollIntoView({
    behavior:"smooth"
  });
}

renderProducts();
</script>

</body>
```

</html>

8.Output Screenshort



43% OFF



Wireless Earbuds

Accessories

★ 4.3 / 5

₹1,999 ~~₹3,499~~

• Product listed

View Details | Compare

Add to Cart 🛒

38% OFF



Fast Charger

Accessories

★ 4.2 / 5

₹799 ~~₹1,299~~

• Product listed

View Details | Compare

Add to Cart 🛒

33% OFF



Smart Watch

Smart Devices

★ 4.4 / 5

₹2,999 ~~₹4,499~~

• Product listed

View Details | Compare

Add to Cart 🛒

38% OFF



Smart Speaker

Smart Devices

★ 4.1 / 5

₹2,499 ~~₹3,999~~

• Product listed

View Details | Compare

Add to Cart 🛒

Product Details & Specifications

Select "View Details" on any product to see its specifications.

📊 Product Comparison

Select "Compare" on product cards to add products to this table. Compare up to 4 products.

Product	Category	Processor / Feature	Memory / Storage	Display	Price	Rating	Action
No products selected. Add products using Compare.							

Clear Comparison

Why Shop at ElectroMart?

🚚 Reliable Delivery

Convenient delivery options for your electronics.

🛡️ Warranty Information

Check the manufacturer's warranty for each product.

💳 Easy Shopping

Browse products and compare specifications in one place.

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